

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26007
- Problem Statement Titles – Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.
- Theme – Smart Automation
- PS Category – Hardware
- Team ID – HW26A06
- Team Name – Team Drashti
- Team Member Name – Pradhumn Joshi (Leader)

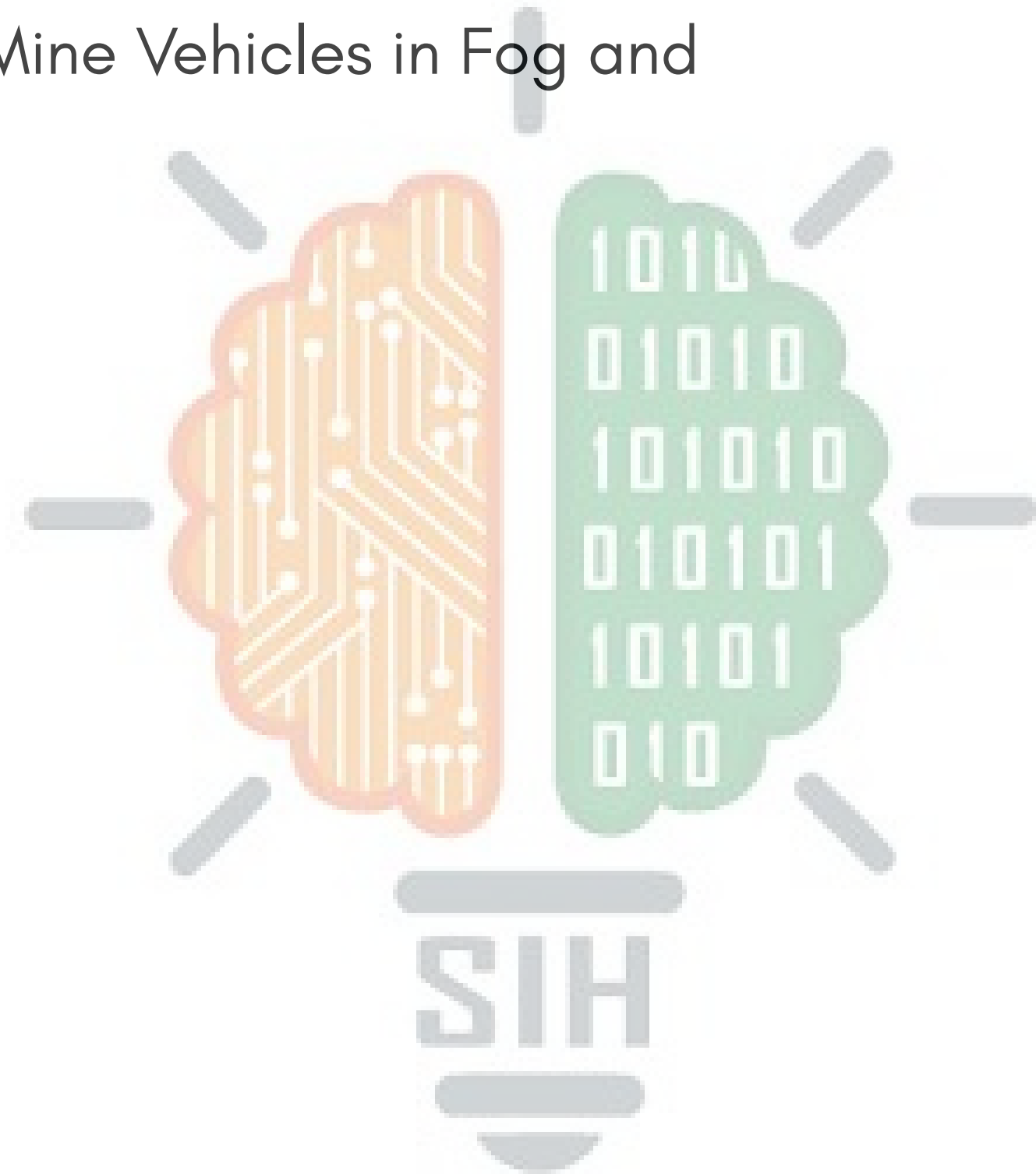
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Impact of ESP32-Based Multi-Sensor Safety System on Call Mines

From Losses Due to Fog - To Profit With Smart Safety



PROBLEM:

- Dense fog reduces visibility to 5-7 meters, increasing accident risk and vehicle stoppages.
- Reduced vehicle movement increases haul cycle time and causes production/profit loss.

SOLUTION:

- An ESP32-based smart safety system monitors the vehicle's surroundings using sensors and a camera.
- Obstacles are detected in real time and the driver receives immediate alerts

Prototype Working:

- Hypersonic Sensors continuously monitor the area in front of the mine vehicle.
- ESP32-CAM provides visual information.
- Distance sensor detects nearby obstacles.
- ESP32 processes the readings in real time.
- Based on distance/risk level, the system activates SAFE, CAUTION or DANGER indication.
- A buzzer provides an instant warning to the driver.

TECHNICAL APPROACH



METHODOLOGY & PROCESS WORKING FLOW

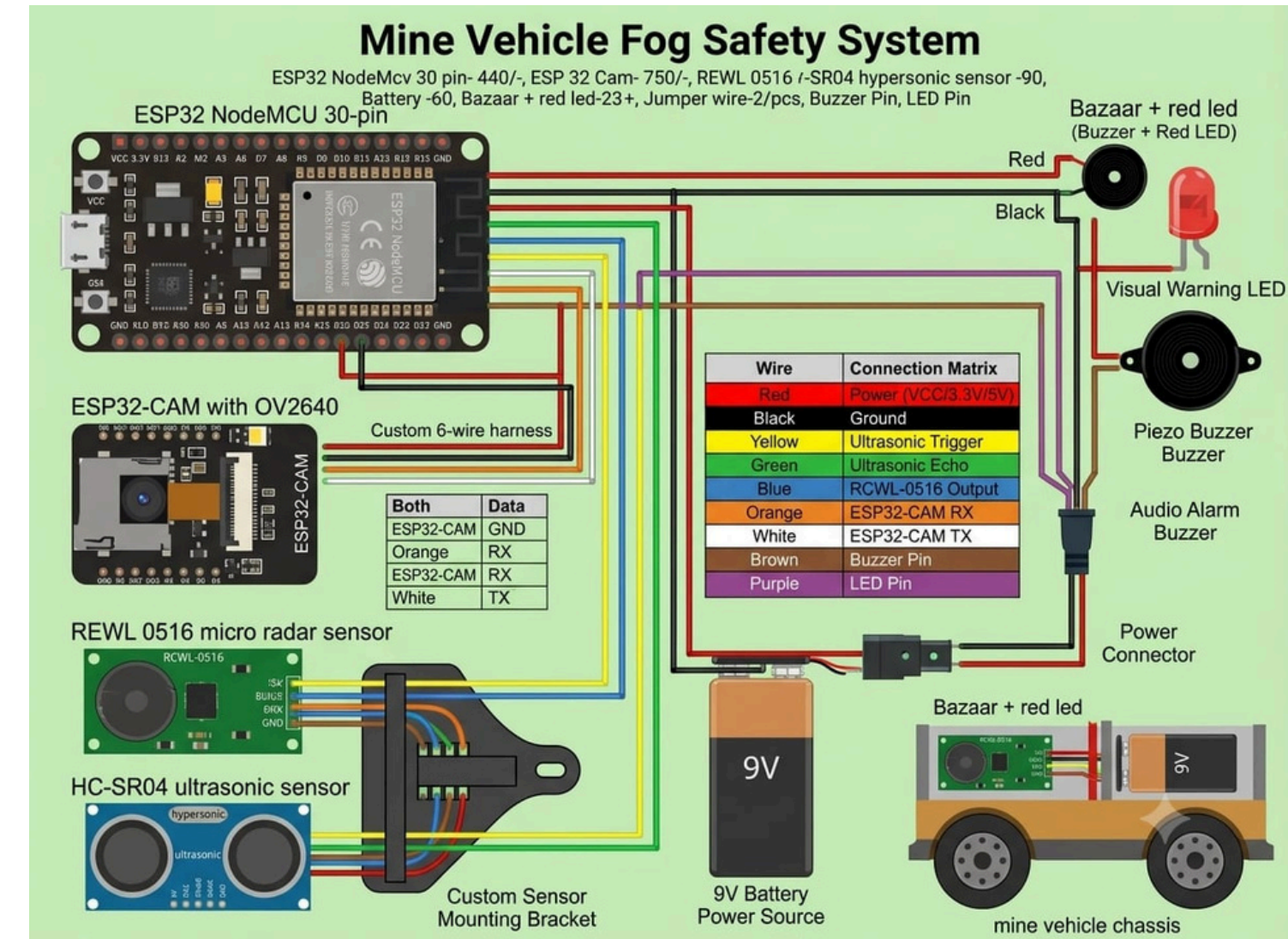


COMPONENT

- ESP32 Board- Main controller
- ESP32-CAM -Camera/visual monitoring
- Radar/Obstacle Sensor-Object detection
- Ultrasonic Sensor-Nearby obstacle ki distance
- Buzzer-Danger warning
- Motor Driver-Motor control
- Gear Motor-Prototype vehicle movement
- Servo Motor-Sensor/camera positioning
- Switches-ON/OFF/control
- 5V Battery
- ESP32 Coding Cable-Programming

SOFTWARE TECHNOLOGIES

- Embedded C/C++ - ESP32 programming through
- Arduino IDE - Microcontroller programming
- ESP32-CAM - Image/video-based monitoring
- IoT/Serial Communication - Sensor data transfer - Monitoring dashboard
- GPS module (optional)
- Key Technologies:
- IoT + Sensors + Computer Vision + Real-Time
- Alert System



FEASIBILITY AND VIABILITY



FEASIBILITY

1



Feasibility of the Idea

- **Low-cost:** ESP32-CAM + sensors are affordable.
- **Real-time:** Detects obstacles and gives instant alerts.
- **Practical:** Works in fog, dust and low visibility.

2



Challenges & Risks

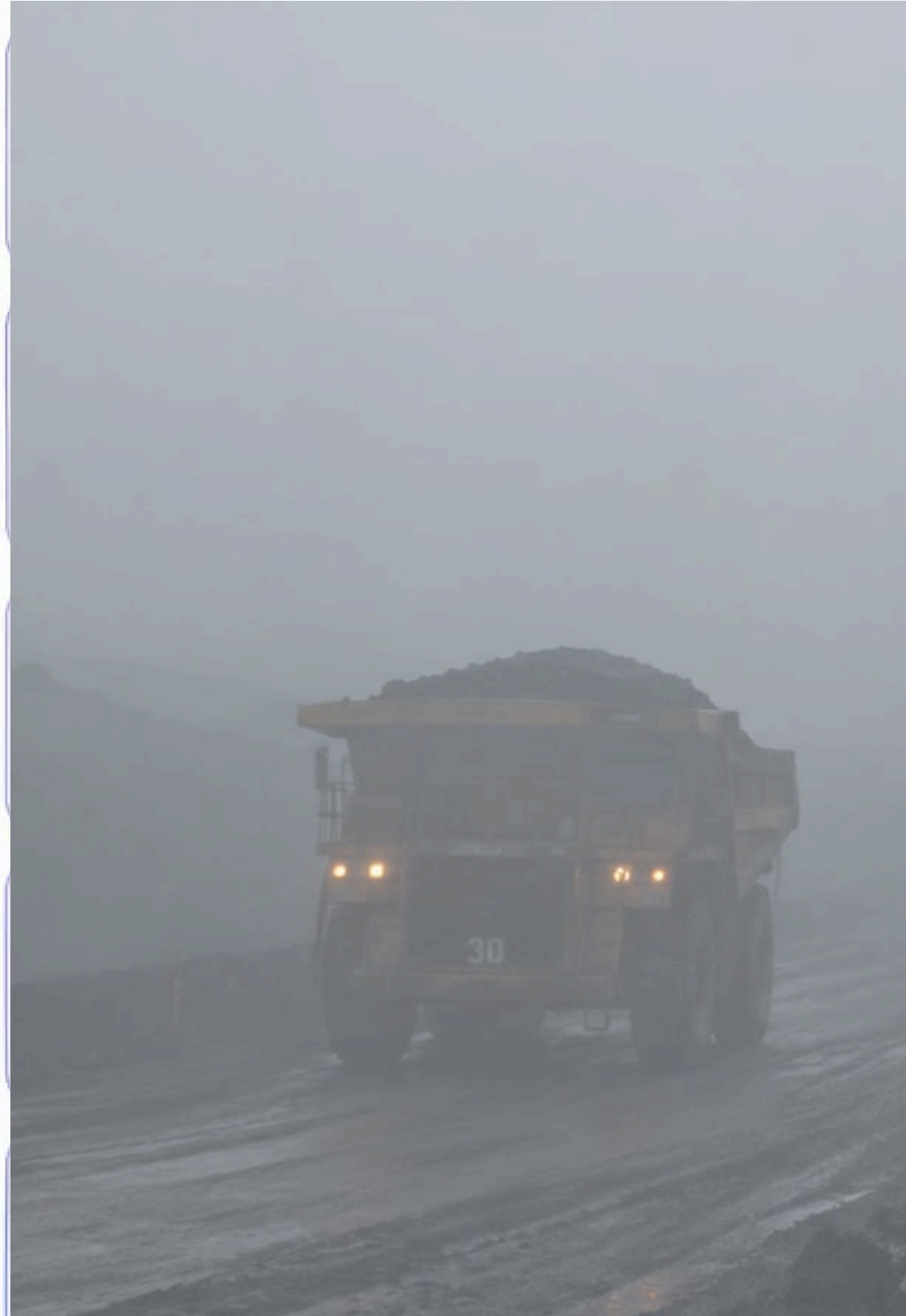
- Heavy fog/dust may affect sensors.
- Harsh mine conditions can damage hardware.
- False alerts may occur.

3



Strategies

- Use multiple sensors for better accuracy.
- Protective casing for dust & vibration.
- Set distance thresholds to reduce false alerts.
- Safe / Caution / Danger warnings for quick driver response.



VIABILITY

1



Affordable

Uses low-cost and easily available components.

2



Scalable

Can be upgraded with industrial-grade sensors for real mines.

3



Reliable

Multi-sensor detection improves safety and accuracy.

4



Useful

Can reduce collision risk and improve driver awareness in fog and dust.

5



Future Scope

Can be integrated with GPS, vehicle tracking and mine control systems.

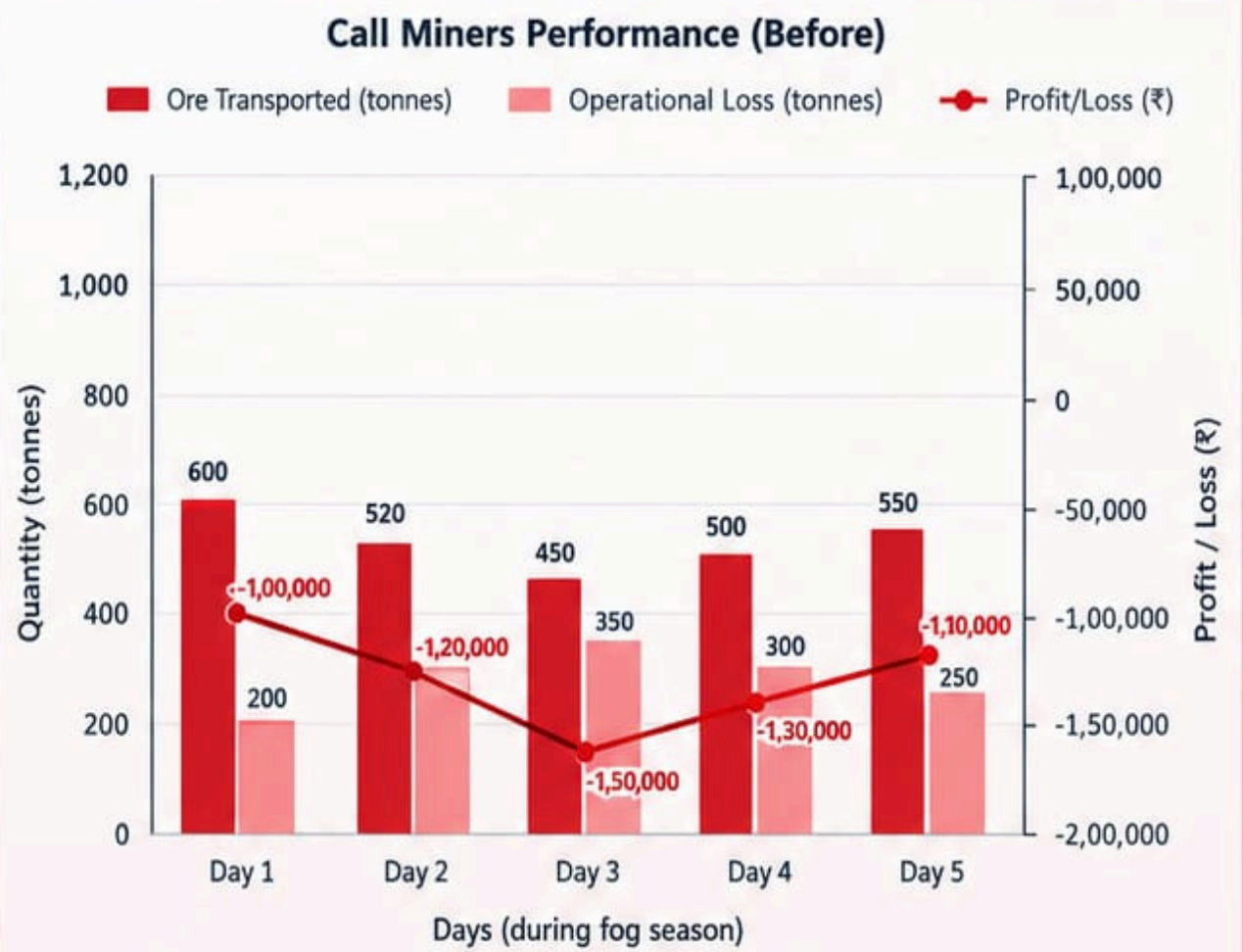
IMPACT AND BENEFIT



- 1 Improved Safety**
Reduces collision risk in fog, dust and rain.
- 2 Faster Response**
Provides early hazard detection and alerts.
- 3 Reduced Accidents**
Helps prevent vehicle-vehicle and vehicle-worker collisions.
- 4 Less Downtime**
Supports safer controlled operation during low visibility.
- 5 Higher Productivity**
Reduces unnecessary operational stoppages.
- 6 Reduced Vehicle Damage**
Prevents avoidable collisions and maintenance costs.
- 7 Smart Automation**
Enables automatic warning and preventive actions.
- 8 Future-ready**
Can be upgraded with AI and autonomous driving features.

Before – Fog Without Safety System

Low visibility, high risk, frequent stoppages → Loss for call miners

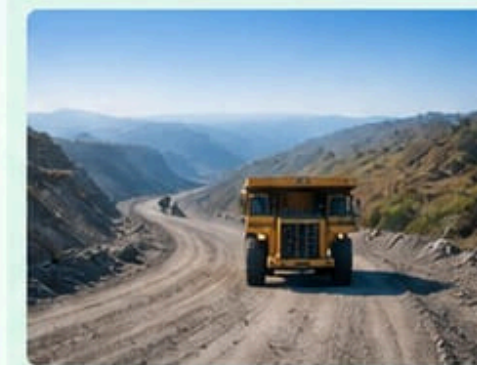
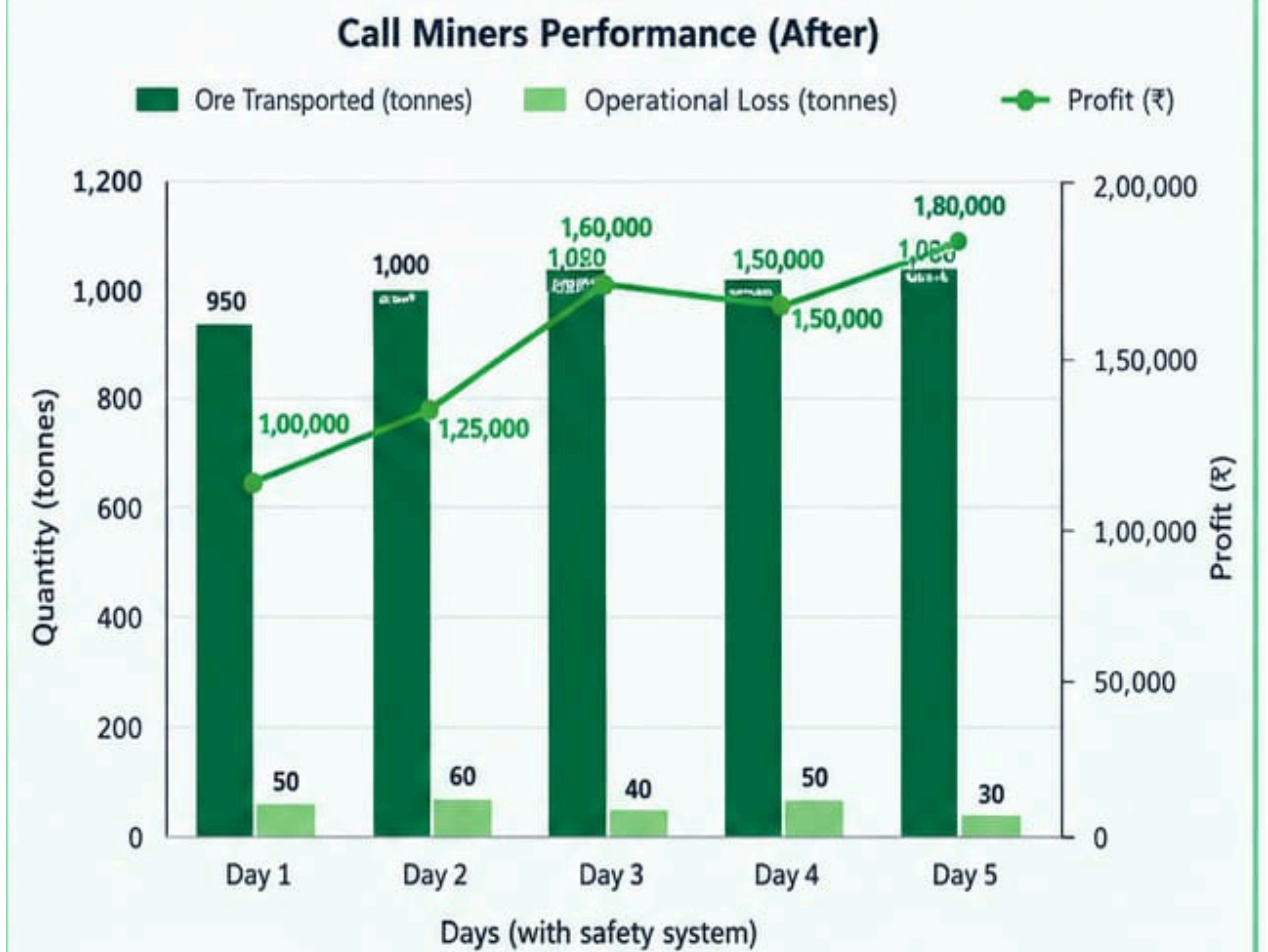


Key Issues Before System

- Visibility reduced to 3–5 m
- Frequent vehicle stoppages
- Higher accident risk
- Increased haul cycle time
- Low fleet utilization
- Production loss and financial loss

After – With ESP32 Multi-Sensor Safety System

Better visibility, early alerts, safer operations → Profit for call miners



Improvements After System

- Better obstacle detection & alerts
- Reduced stoppages
- Safer vehicle movement
- Shorter haul cycle time
- Higher fleet utilization
- Increased production & profit

RESEARCH AND REFERENCE



1. Espressif Systems – ESP32 Documentation

https://documentation.espressif.com/esp32_datasheet_en.html?utm_source=chatgpt.com

For ESP32 specifications, GPIOs, power and technical characteristics.

2. Espressif Documentation

Espressif Systems –

https://www.espressif.com/en/support/documents/technical-documents/esp32-?utm_source=chatgpt.com

For ESP32 hardware and development documentation.

3. Arduino Documentation

https://docs.arduino.cc/?utm_source=chatgpt.com

For Arduino IDE, programming and microcontroller development.

4. Arduino ESP32 Core–

https://docs.espressif.com/projects/arduino-esp32/en/latest/?utm_source=chatgpt.com

For programming ESP32 using the Arduino framework.

5. Directorate General of Mines Safety (DGMS), India–

https://dgms.gov.in/?utm_source=chatgpt.com

For Indian mine-safety regulations and guidance.

6. Ministry of Mines, Government of

India <https://mines.gov.in/webportal/home>

For information and background related to mining in India

7. HC-SR04 & RCWL-0516 Sensor Documentation

For sensor specifications, working principles, and technical details.